

Media Pipeline & Self-Hosted Storage

Capture everything the cluster and your dev box generate — music, images, video, PDFs, 3D meshes — into one durable store, then browse, search, and stream it with the right self-hosted app for each.

The one-line mental model

Producers push to MinIO (one source of truth) → **small syncs fan out** to purpose-built apps. Producers only ever learn "*push to S3*" — they never learn how each app works.



Immich **CLIP**

Images + video library with semantic search & face recognition. Your CLIP model, set up for free.



Navidrome

Dedicated music player for **acstep** songs. Subsonic clients on iOS & Apple TV.



Paperless-ngx

PDF library: OCR, full-text search, tags. For your debugging & HyperFrames PDFs.

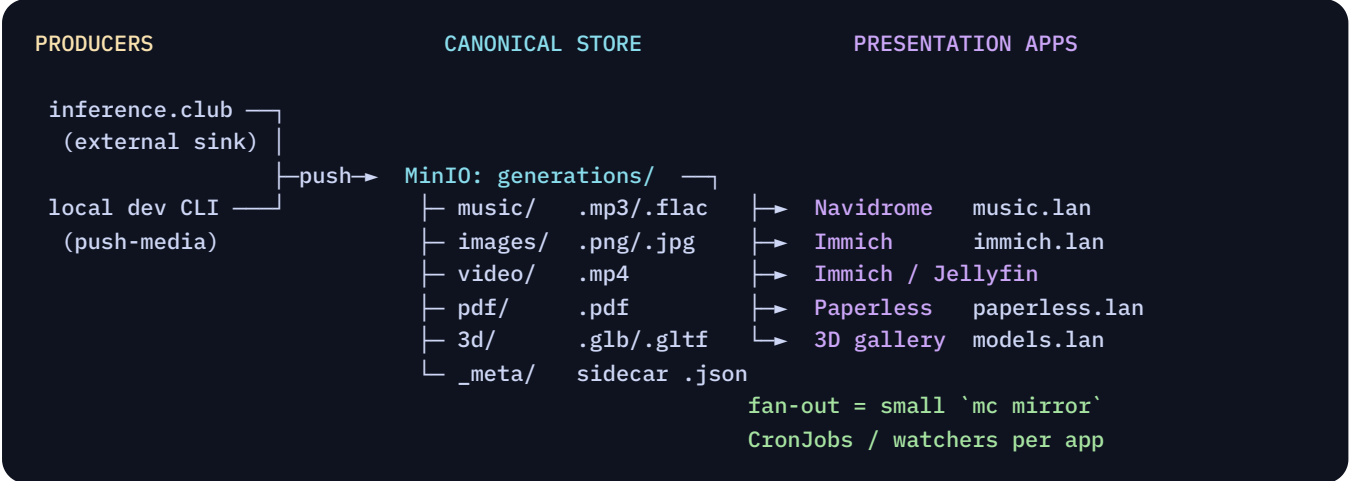


3D gallery

`<model-viewer>` gallery for **trellis2** meshes, reading straight from MinIO.

Decisions locked (2026-06-29): MinIO landing zone + fan-out · build all four apps on **a2** · Navidrome (dedicated) for music · backups **sketched but deferred**.

1 Architecture — MinIO landing zone + fan-out



1 MinIO is the source of truth

Every artifact lands first as `generations/<type>/<id>.<ext>` with an optional `_meta/` JSON sidecar. Back up one bucket → back up everything.

2 Producers only speak S3

inference.club & the local CLI need only an endpoint, bucket, prefix & creds. They never learn how Immich or Navidrome work.

3 Fan-out is a cluster concern

Each app gets a tiny `mc mirror` sync into its preferred shape. Swap or add apps without touching producers.

4 Keep it boring

MinIO ships `mc`; v1 needs **no custom code** — a CronJob or `mc mirror --watch` sidecar per destination.

Fan-out per app

DESTINATION	SYNC	APP BEHAVIOR
Navidrome	<code>mc mirror .../music/ → /music</code>	auto-scans library
Paperless	<code>mc mirror .../pdf/ → consume/</code>	consumes + OCRs, then removes
Immich	<code>mc mirror .../{images,video}/ → /import</code>	External Library (read-only) indexes folder
3D gallery	<i>none</i> — reads MinIO directly	lists <code>3d/</code> , renders inline via presigned URLs

2 The four apps (build-now)



Immich — images + video + CLIP highest value

Library with a great mobile app, **CLIP semantic search**, face recognition, and a real **API + CLI**. This is also how you stand up a CLIP model — its ML container does it out of the box.

Where: a2 (server + ML + Postgres + Redis). **CLIP/GPU:** a2's GPU is taken by flux2 → run ML **CPU-first**, share GPU later if needed. **Alt:** PhotoPrism — weaker search, no clean ingest API.



Navidrome — music

Tiny Subsonic-compatible server for **acestep** songs; iOS/Apple TV clients (Amperfy, play:Sub).

Where: a2, one container + SQLite. **Alt:** reuse Jellyfin music (simpler, worse UX).



Paperless-ngx — PDFs

Drop a PDF → OCR, full-text search, tags, web viewer. For debugging & HyperFrames PDFs.

Where: a2 (web + Postgres + Redis). **Alt:** Stirling-PDF = tools, a complement.



3D gallery — trellis2 meshes small custom build

No mature "Immich for 3D," so: a tiny static app that lists **generations/3d/** and renders each **.glb/.gltf** inline with Google's **<model-viewer>** (orbit/zoom in-browser).

Where: a2, small image via CI → Harbor. Reads MinIO via presigned URLs — no separate storage. **Auth:** none → keep LAN-only for now.

3 The producer contract

Local dev — push-media CLI

```
push-media song.mp3 --type music
push-media debug.pdf --type pdf
push-media mesh.glb --type 3d
```

Wraps `mc cp`; uploads to `generations/<type>/` + writes a `_meta/` sidecar.

inference.club — "external sink"

An optional config block

(`S3_ENDPOINT/BUCKET/PREFIX/KEY/SECRET`).

When set, every generation is mirrored to your

MinIO using the same contract. inference.club stays authoritative; the cluster gets a local copy to present.

(Lives in that repo.)

4 Phased build order

0 Contract do first

MinIO `generations/` buckets + scoped creds + `push-media` CLI. The foundation everything rides on.

1 Immich build now

Images/video library + CLIP search on a2. Highest value; also stands up CLIP.

2 Navidrome + Paperless build now

Music player + PDF library on a2. Two quick, lightweight wins.

3 3D gallery build now

model-viewer gallery reading MinIO. Small custom image via CI → Harbor.

4 inference.club sink build now

"External sink" S3 mirror in the product (inference.club repo).

5 Backups deferred

restic → Backblaze B2 + a real restore drill. Documented now, built later.

5 Storage & backup — sketched, deferred

The cluster has **no backups today** (flagged risk). MinIO-as-truth makes it tractable: protect a short list and you've protected everything.

Back up: MinIO `generations/` + app DBs (Immich PG, Paperless PG+media, Navidrome SQLite). 3D gallery is stateless. **How (later):** restic → offsite **Backblaze B2** for 3-2-1; Longhorn/second-node for on-LAN mobility. **Note:** the NTFS d/e drives are read-only — not valid backup targets. A backup you've never restored isn't a backup.

6 Cross-cutting

TLS/DNS new `*.lan` hosts → `scripts/lan-certs.sh` **Homepage** ingress annotations, a "Media" group
Tailscale expose only Immich (+ maybe Navidrome) remotely **Auth** apps self-auth; 3D gallery LAN-only until SSO **GPU** only Immich ML wants one → CPU-first on a2.